

Phase 5: Splitting the Atom

Nuclear Fission, Mass Defect, and the Ultimate Boiler

The Virtual Professor

The Professor's Intuition: The Ultimate Energy Density

When we burn coal, we are just breaking weak chemical bonds between carbon and oxygen. It releases a tiny spark of energy (measured in electron-Volts). But the nucleus of an atom is held together by the Strong Nuclear Force—the most powerful force in the universe.

If we can crack a heavy nucleus (like Uranium) in half, a tiny fraction of its physical mass literally vanishes. Where does it go? Albert Einstein gave us the answer: $E = mc^2$. Because c (the speed of light) is such a massive number, even a microscopic speck of lost mass converts into a cataclysmic amount of heat. **Just 1 kg of Uranium-235 produces the same heat as burning 3,000,000 kg of coal!** A nuclear power plant is just a steam plant that uses quantum physics instead of a coal fire to boil the water.

1 The Physics of Fission

You cannot just smash atoms together. To split Uranium-235, you must shoot a **neutron** at it. When the U-235 nucleus absorbs the neutron, it becomes violently unstable and snaps into two smaller elements (like Barium and Krypton).

But here is the magic: the split also shoots out **3 new free neutrons**. If these 3 neutrons hit 3 more Uranium atoms, they split, releasing 9 neutrons... then 27... then 81. This is a **Chain Reaction**. If uncontrolled, it is an atomic bomb. If controlled, it is a nuclear power plant.

2 Anatomy of the Reactor Core

(Frequently tested: Functions of components - 2022 Q1, 2023 Q2)

To keep the chain reaction from melting the city, we use very specific engineering controls inside the reactor core:

1. **The Moderator (The Speed Bump):** When a uranium atom splits, the new neutrons shoot out at massive speeds ("fast neutrons"). Paradoxically, fast neutrons usually bounce off other uranium atoms without splitting them. We must slow them down to "thermal" speeds. The Moderator (often water or graphite) sits between the fuel rods. The neutrons crash into the moderator molecules, slowing down just enough to trigger the next fission.
2. **Control Rods (The Brakes):** Made of Cadmium or Boron, these rods act like neutron sponges. If the reactor gets too hot, operators lower the rods deep into the core. They absorb the bouncing neutrons, starving the chain reaction and cooling the reactor. Drop them all the way in, and the reactor shuts down instantly (a "SCRAM").
3. **The Reflector (The Mirror):** Neutrons at the edge of the core try to escape into the walls. We surround the core with a Reflector (like Beryllium) to bounce the escaping neutrons back into the fuel, saving money and improving efficiency.

3 The Big Two: BWR vs. PWR

Every semester, you are asked to draw and explain one of these two reactors. The fundamental difference between them is **how they handle the radioactive water**.

3.1 1. Boiling Water Reactor (BWR)

(Appeared in 2023 Q6 for 10 marks)

Concept: This is the simplest design (a "Direct Cycle"). The water inside the reactor acts as both the coolant and the moderator. The nuclear heat boils the water **directly inside the reactor vessel**. The steam goes straight to the turbine.

- **Pro:** Cheaper and simpler to build (only one loop).
- **Con:** The water touches the radioactive core, becomes slightly radioactive, and carries that radiation straight into the turbine building. The turbine becomes a radiation zone, making maintenance difficult.

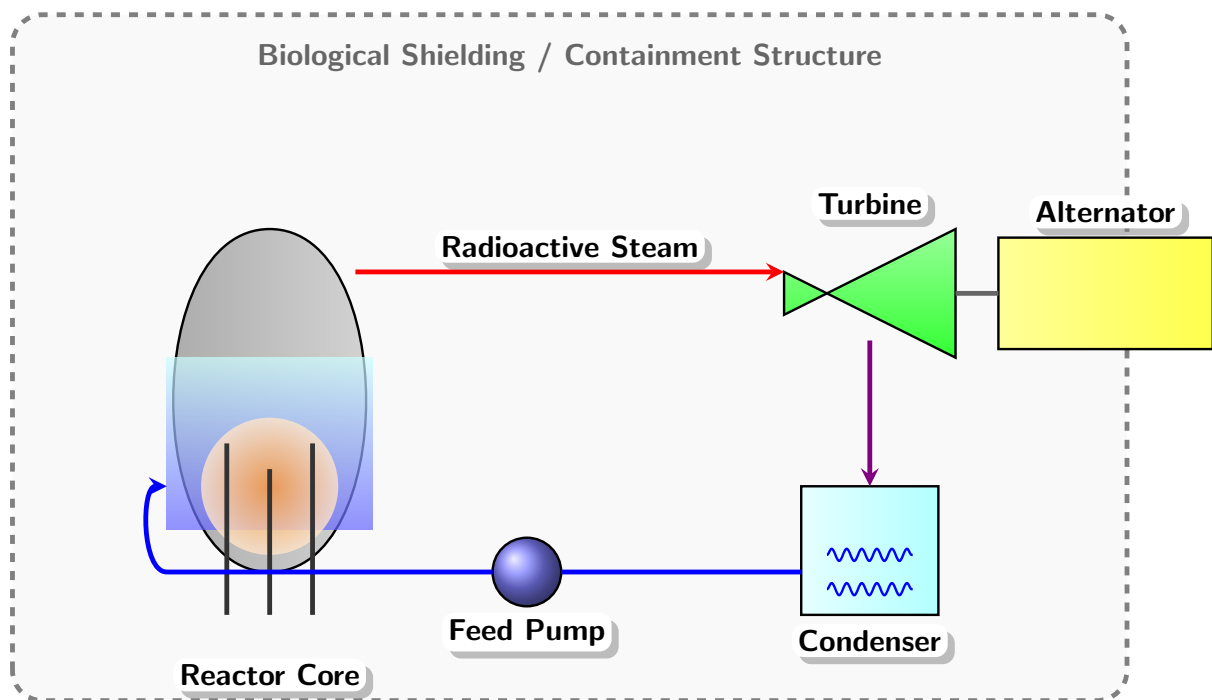


Figure 1: Boiling Water Reactor (BWR) - Direct Cycle

3.2 2. Pressurized Water Reactor (PWR)

(Appeared in 2025 Q3 and 2022 Q2c for 10 marks)

Concept: This is a "Two-Loop" system. We do **not** want the water inside the reactor to boil. So, we add a **Pressurizer** to squeeze the Primary Loop to a crushing 150 atmospheres. At this pressure, water stays liquid even at 300°C. This super-hot, radioactive liquid water is pumped to a separate Heat Exchanger (Steam Generator). Here, its heat boils the water in the *Secondary Loop*.

- **Pro:** The radioactive primary water never touches the secondary water. The steam going to the turbine is 100% clean and safe to work around.
- **Con:** Much more expensive and complex.

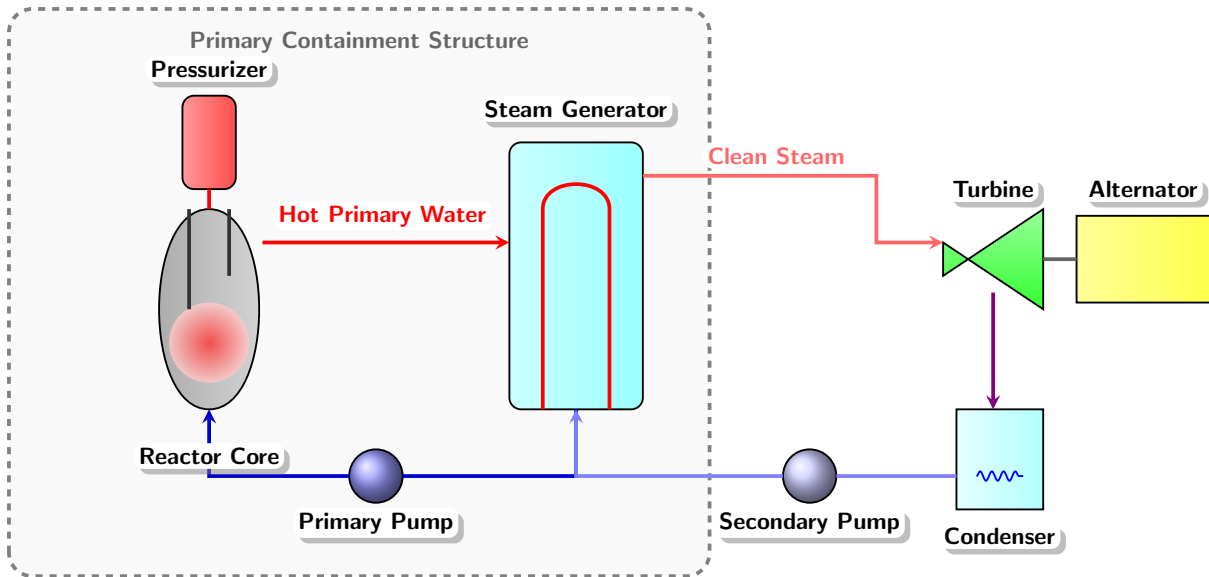


Figure 2: Pressurized Water Reactor (PWR) - Dual Loop System

4 The Fast Breeder Reactor (FBR)

(Appeared as Short Note in 2024 Q5b)

Normal reactors only run on U-235, which is incredibly rare (only 0.7% of natural uranium). The rest is U-238, which is useless in a normal reactor.

A **Fast Breeder Reactor** does the impossible: it creates more fuel than it consumes.

- **No Moderator:** It uses "fast neutrons" (no water or graphite to slow them down).
- **The Breeding:** When fast neutrons hit the useless U-238 blanket surrounding the core, they transmute it into **Plutonium-239**, a brand new, highly powerful nuclear fuel.
- **The Coolant:** Because we cannot use water (since water is a moderator and would slow the neutrons), FBRs must use **Liquid Sodium** to cool the intensely hot core.

5 The Aftermath: Waste & Location

(Appeared in 2022 Q1vi)

Waste Management: Radioactive waste does not just disappear; it stays dangerous for thousands of years.

- **Low-Level Waste:** (Contaminated gloves, tools, cooling pipes). These are compacted, sealed in concrete/steel drums, and buried in shallow, stable geological trenches.
- **High-Level Waste:** (Spent fuel rods). These are incredibly hot and radioactive. They are first kept in deep water cooling pools on-site for several years. Once slightly cooled, they are vitrified (turned into solid glass blocks to prevent leaking) and sealed in massive lead-steel casks for deep underground permanent burial.